

Experiment 9

1. Aim

Perform OS hardening on a Linux system by disabling unused services, configuring secure SSH settings etc.

2. Hardware / Software Required

- Ubuntu/Debian/Kali Linux
- Terminal
- OpenSSH Server
- systemctl
- ufw
- Text editor such as nano or vim

3. Theory

OS hardening is the process of securing an operating system by reducing unnecessary functionality, restricting access, and configuring security controls. The main objective is to reduce the system's **attack surface** and make exploitation more difficult.

1. Disabling Unused Services

Linux systems may have services running in the background that are not required. Every unnecessary service can potentially provide an additional entry point for an attacker.

Hardening involves:

- Identifying running services
- Determining which services are actually required
- Stopping unnecessary services
- Preventing unnecessary services from starting automatically

2. Secure SSH Configuration

SSH (Secure Shell) provides encrypted remote access to a Linux system. Incorrect SSH configuration can increase the risk of unauthorized access.

Common SSH hardening practices include:

- Disable direct root login
- Prefer key-based authentication
- Disable password authentication when key-based access is properly configured

- Use appropriate SSH access restrictions
- Keep SSH software updated
- Avoid unnecessary exposure of SSH to untrusted networks

The main configuration file is: `/etc/ssh/sshd_config`

3. Firewall Configuration

A firewall restricts unwanted network connections. During hardening, only required ports and services should be accessible.

For example:

- SSH → port 22
- HTTP → port 80
- HTTPS → port 443

All unnecessary incoming connections should generally be blocked.

4. User and Privilege Management

Linux follows the principle of **least privilege**, where users should receive only the permissions necessary to perform their tasks.

Hardening may include:

- Removing unnecessary accounts
- Using strong passwords
- Limiting sudo privileges
- Avoiding routine work as root
- Disabling inactive accounts

5. System Updates

Security vulnerabilities can exist in outdated operating-system packages and services. Regularly installing security updates helps protect against known vulnerabilities.

4. Procedure

1. Log in to the Linux system with an administrative account.
2. Identify the operating system version and review the currently running services.
3. Identify services that are unnecessary for the system's intended purpose.
4. Stop unnecessary services and disable them from automatically starting at boot.
5. Review user accounts and remove or disable unnecessary accounts.
6. Review the SSH configuration and make the necessary security changes.
7. Disable direct root login through SSH.
8. Configure SSH to use stronger authentication methods, preferably SSH keys.

9. Configure the firewall to allow only required network services.
10. Apply available system and security updates.
11. Verify that unnecessary services are no longer running.
12. Verify the SSH configuration and firewall rules.
13. Reboot the system if required and confirm that the security configurations remain active.
14. Record the observations and results.

Command	Purpose
systemctl list-units --type=service --state=running	Lists currently running services
systemctl status <service>	Checks the status of a service
sudo systemctl stop <service>	Stops a running service
sudo systemctl disable <service>	Prevents a service from starting at boot
sudo systemctl enable <service>	Enables a service at boot
sudo systemctl restart ssh	Restarts the SSH service
sudo nano /etc/ssh/sshd_config	Opens the SSH configuration file
sudo sshd -t	Tests the SSH configuration for syntax errors
sudo ufw status	Displays firewall status
sudo ufw allow 22/tcp	Allows SSH traffic
sudo ufw enable	Enables UFW
sudo ufw status numbered	Displays numbered firewall rules
sudo apt update	Updates the package information
sudo apt upgrade	Installs available package updates

5. Output

```
kali@kali: ~
Session Actions Edit View Help
(kali@kali)-[~]
$ systemctl list-units --type=service --state=running
UNIT                                LOAD    ACTIVE SUB    DESCRIPTION
accounts-daemon.service             loaded active running Accounts Service
colord.service                       loaded active running Manage, Install and Generate Color Profiles
cron.service                        loaded active running Regular background program processing daemon
dbus.service                        loaded active running D-Bus System Message Bus
getty@tty1.service                  loaded active running Getty on tty1
haveged.service                     loaded active running Entropy Daemon based on the HAVEGE algorithm
lightdm.service                     loaded active running Light Display Manager
ModemManager.service                loaded active running Modem Manager
NetworkManager.service              loaded active running Network Manager
polkit.service                       loaded active running Authorization Manager
rtkit-daemon.service                loaded active running RealtimeKit Scheduling Policy Service
ssh.service                          loaded active running OpenBSD Secure Shell server
systemd-journald.service             loaded active running Journal Service
systemd-logind.service               loaded active running User Login Management
systemd-udev.service                 loaded active running Rule-based Manager for Device Events and Files
systemd-userdbd.service              loaded active running User Database Manager
udisks2.service                     loaded active running Disk Manager
upower.service                       loaded active running Daemon for power management
user@1000.service                   loaded active running User Manager for UID 1000
virtualbox-guest-utils.service        loaded active running Virtualbox guest utils

Legend: LOAD → Reflects whether the unit definition was properly loaded.
ACTIVE → The high-level unit activation state, i.e. generalization of SUB.
SUB → The low-level unit activation state, values depend on unit type.

20 loaded units listed.

(kali@kali)-[~]
$ systemctl status cron.service
● cron.service - Regular background program processing daemon
   Loaded: loaded (/usr/lib/systemd/system/cron.service; enabled; preset: enabled)
   Active: active (running) since Sun 2026-09-06 09:50:18 EDT; 21min ago
   Invocation: 2267bf9087ce4833a7a0c4f1b0b3ebdc
     Docs: man:cron(8)
   Main PID: 507 (cron)
     Tasks: 1 (limit: 4450)
   Memory: 524K (peak: 2M)
     CPU: 160ms
   CGroup: /system.slice/cron.service
           └─507 /usr/sbin/cron -f

Sep 06 09:50:18 kali cron[507]: (CRON) INFO (Running @reboot jobs)
Sep 06 09:55:01 kali CRON[4757]: pam_unix(cron:account): expired password for user root (password aged)
Sep 06 09:55:01 kali CRON[4757]: Authentication token is no longer valid; new one required
Sep 06 09:55:01 kali cron[4757]: Authentication token is no longer valid; new one required
Sep 06 10:05:01 kali CRON[10004]: pam_unix(cron:account): expired password for user root (password aged)
Sep 06 10:05:01 kali CRON[10004]: Authentication token is no longer valid; new one required
Sep 06 10:05:01 kali cron[10004]: Authentication token is no longer valid; new one required
Sep 06 10:09:01 kali CRON[11926]: pam_unix(cron:account): expired password for user root (password aged)
Sep 06 10:09:01 kali cron[11926]: Authentication token is no longer valid; new one required
Sep 06 10:09:01 kali CRON[11926]: Authentication token is no longer valid; new one required

(kali@kali)-[~]
```

```
kali@kali: ~
Session Actions Edit View Help
Sep 06 10:09:01 kali CRON[11926]: Authentication token is no longer valid; new one required

(kali@kali)-[~]
$ sudo systemctl stop <service>
zsh: parse error near `\'

(kali@kali)-[~]
$ sudo systemctl stop ssh.service
[sudo] password for kali:

(kali@kali)-[~]
$ sudo systemctl restart ssh

(kali@kali)-[~]
$ systemctl list-units --type=service --state=running
UNIT                                LOAD    ACTIVE SUB    DESCRIPTION
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(kali@kali)-[~]
$
```

```
GNU nano 9.0 /etc/ssh/sshd_config
# This is the sshd server system-wide configuration file. See
# sshd_config(5) for more information.

# This sshd was compiled with PATH=/usr/local/bin:/usr/bin:/bin:/usr/games

# The strategy used for options in the default sshd_config shipped with
# OpenSSH is to specify options with their default value where
# possible, but leave them commented. Uncommented options override the
# default value.

Include /etc/ssh/sshd_config.d/*.conf

#Port 22
#AddressFamily any
#ListenAddress 0.0.0.0
#ListenAddress ::

#HostKey /etc/ssh/ssh_host_rsa_key
#HostKey /etc/ssh/ssh_host_ecdsa_key
#HostKey /etc/ssh/ssh_host_ed25519_key

# Ciphers and keying
#RekeyLimit default none

# Logging
#SyslogFacility AUTH
#LogLevel INFO

# Authentication:

#LoginGraceTime 2m
#PermitRootLogin prohibit-password
#StrictModes yes
#MaxAuthTries 6
#MaxSessions 10

#PubkeyAuthentication yes

# Expect .ssh/authorized_keys2 to be disregarded by default in future.
#AuthorizedKeysFile .ssh/authorized_keys .ssh/authorized_keys2
```

```
File Machine View Input Devices Help
kali@kali: ~
20 loaded units listed.
(kali@kali)-[~]
$ sudo nano /etc/ssh/sshd_config
(kali@kali)-[~]
$ sudo sshd -t
(kali@kali)-[~]
$ sudo ufw status
Status: inactive
(kali@kali)-[~]
$ sudo ufw allow 22/tcp
Skipping adding existing rule
Skipping adding existing rule (v6)
(kali@kali)-[~]
$ sudo ufw enable
Firewall is active and enabled on system startup
(kali@kali)-[~]
$ sudo ufw status numbered
Status: active

To Action From
--
[ 1] 22/tcp ALLOW IN Anywhere
[ 2] 80/tcp ALLOW IN Anywhere
[ 3] 22/tcp (v6) ALLOW IN Anywhere (v6)
[ 4] 80/tcp (v6) ALLOW IN Anywhere (v6)
```

Conclusion

Thus, **OS hardening was successfully performed on a Linux system.** Unused services were identified and disabled, SSH was configured with security in mind, unnecessary network access was restricted using firewall rules, and system updates were applied. These measures reduce the system's attack surface and provide better protection against unauthorized access and common security threats.